

Problem Chosen
A

MCM/ICM
Summary Sheet

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A Hot Bath: Conserving Water Without Losing Uniformity

Summary

Maintaining a warm bath is a coupled problem of heat loss, replenishment and transport. A hot inlet can improve the mean temperature while leaving distant water cool and sending useful heat directly to the overflow. We therefore minimize added water subject to explicit limits on every modeled cell, rather than optimizing an average alone.

We prove a coast-then-hold policy optimal for a well-mixed bath, then use a three-dimensional finite-volume thermal network with coefficients derived from textbook correlations and tied to a published immersion study where one exists (Section 2). The best constant rate is compared with an optimized piecewise-constant schedule; neither is proved optimal among all controls.

In a 164.25 L water-volume scenario lasting 30 minutes, with an initial temperature of 40°C and a 39°C lower limit, the best constant-rate policy adds **24.14 L** at 0.805 L/min from the start, and a 12-segment schedule found by constrained optimization needs **19.35 L**, 20% less. The well-mixed optimum is 16.01 L, while an independent energy argument gives a 15.41 L lower bound for the spatial problem. A finer mesh changes the constant-rate result by 0.11%, and the constrained temperatures agree across three meshes.

The answer depends strongly on the surface and body loss coefficients. Across 64 Sobol draws over the scenario ranges of Section 2.1, 41 had an accepted constant-rate policy, with required water from 12 to 31 L (5th–95th percentile, median 22 L); whether a draw has an accepted policy at all depends mainly on the mixing diffusivity. Weak mixing yields no accepted candidate under the stated temperature limits; stronger mixing saves water, but added surface loss can reverse that benefit. In the assumed foam scenario replenishment falls to 10.4 L. A tolerance sweep puts each extra 0.5°C of allowed fall at about 11 L until a 2.5°C fall needs none, and vertical stratification adds little water and uses only part of the spread margin. These are conditional calculations, not measurements or product claims.

Independent integration, analytic limits, energy accounting and a between-samples envelope support the results: 17 model checks and 2 for the schedule. A one-page user guide and the limitations follow; AI assistance is disclosed in the references and appended report.

Keywords: thermal network; energy balance; lower bound; optimal control; sensitivity analysis

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1 Define the decision before optimizing

The task is to preserve both warmth and spatial uniformity in an overflowing, unheated tub, and to examine geometry, the bather and motion, and a bubble-bath layer [1]. The report separates physical requirements from preference assumptions. There is no supplied temperature record or measured heat-transfer coefficient to fit.

The decision variables are an inlet flow rate and the time at which a constant trickle begins. The tub is already full: added water displaces an equal volume through the overflow. The horizon is 1,800 s. Our baseline requires every cell average to stay at or above 39°C, and requires the cell averages outside a 0.15 m jet-mixing zone around the inlet to stay at or below 41°C and within an instantaneous spread of 1.5°C. These choices operationalize comfort; they are not medical limits or numbers specified by the problem, and the jet zone is an assumption justified in Section 2.1.

$$J = 1000 \int_0^{t_f} q(t) dt \quad (1)$$

$$T_i(t) \geq T_{\min} \quad (\forall i), \quad T_i(t) \leq T_{\max}, \quad \max_{i \in \Omega} T_i(t) - \min_{i \in \Omega} T_i(t) \leq \Delta \quad (i \in \Omega) \quad (2)$$

Here Ω is the set of cells outside the inlet jet zone, q is in m³/s and J is in litres. We prioritize the least water within the constraints, rather than assigning arbitrary weights to unlike units. Tightening the temperature tolerance is a separate scenario. Zero flow is admitted: if the initial stored heat suffices, using no added water is globally water-minimal.

Table 1. Where each requirement is answered

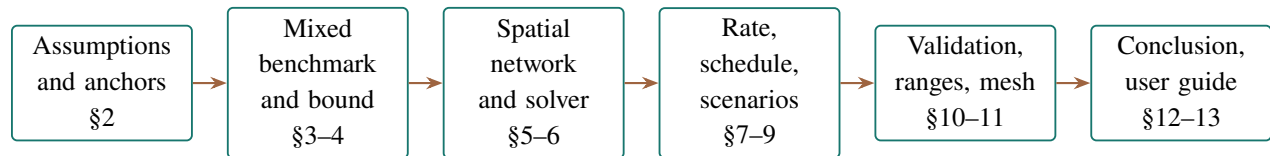
Requirement	Where answered
Temperature in space and time	Sections 3, 5–7; Figures 2–3
Water-use strategy and its scope	Sections 4, 6–7 and 12
Tub/body geometry, size and temperature	Section 8
Motion and bubble-bath additive	Section 9
Validation and sensitivity	Sections 8–11
One-page non-technical explanation	Section 13

Spatial temperature means a control-volume average. It does not bound the unresolved temperature of a faucet jet or a skin-contact film. This distinction determines which practical conclusions the simulation can support.

Modeling options. A lumped model (Newton cooling) keeps one temperature and cannot see a cold corner. A flow solver resolves velocity but needs a turbulence closure and a faucet jet that nothing available here can constrain. A finite-volume thermal network sits between them: it keeps conservation exact, resolves where heat is lost and delivered, and makes mixing an explicit, testable parameter [5]. We use the lumped model as a proved benchmark (Sections 3–4) and the network for the decision (Sections 5–11).

Table 2. Modeling options compared

Option	Resolves	Evidence it needs	Role here
Lumped (Newton)	Mean temperature	Two loss coefficients	Proved benchmark
Thermal network	Cells, heat paths, overflow	Loss, body and mixing coefficients	Decision model
Flow solver	Velocity and buoyancy	Turbulence closure, jet data	Not used

**Figure 1.** Roadmap: how the sections build from assumptions to a recommendation.

2 Physical assumptions and scenario inputs

Each assumption below is paired with its reason and, where it matters, the scenario that tests it. Coefficients cannot be identified from the problem statement alone, so they come from the anchors in Section 2.1.

Table 3. Baseline inputs

Quantity	Baseline / units	Status
Tub $L \times W \times H$	$1.50 \times 0.65 \times 0.23$ m	Assumed geometry
Displaced body volume / area	$0.060 \text{ m}^3 / 1.15 \text{ m}^2$	Assumed
Water density / heat capacity	$1000 \text{ kg/m}^3 / 4180 \text{ J/(kg K)}$	Rounded constants
Air / skin / inlet temperature	$22 / 34 / 50^\circ\text{C}$	Fixed reservoirs
Surface / wall / body coefficient	$25 / 6.5 / 25 \text{ W/(m}^2 \text{ K)}$	Correlation-based (next page)
Mixing diffusivity D	$0.001 \text{ m}^2/\text{s}$	Uncalibrated closure; range tested
Initial / lower / upper limit	$40 / 39 / 41^\circ\text{C}$	Preference scenario
Time / allowed spread	$1800 \text{ s} / 1.5^\circ\text{C}$	Preference scenario

A1. Constant density and heat capacity. Water properties change by under 1% between 39 and 41°C , and the constants are rounded values, not an evaluated IAPWS table [3]. *Reason:* the resulting capacity error is small next to the loss coefficients.

A2. Heat exchange proportional to temperature difference. Surface, shell and body exchange follow Newton's law with effective coefficients [2]. *Reason:* the temperature span is narrow, so linearization is acceptable. The surface coefficient aggregates convection, radiation and evaporation, so evaporation is not added twice; evaporative volume change is neglected. The evaporative part is linearized about the room temperature; an independent review estimated that this

understates its sensitivity near 40°C by several percent (about 7% on the total), which is small next to the tested range.

A3. Skin held at a fixed temperature. *Reason:* thermoregulation is outside the problem; skin temperature is varied (32 and 36°C) instead of resolved.

A4. Inlet-to-overflow surface stream. The hot water travels along a prescribed surface path to the overflow, so short circuit is possible. The stream runs along the row of cells nearest the centre line (half a cell off it on meshes with an even number of rows). *Reason:* a different inlet geometry would need its own transport network and evidence.

A5. Mixing as one effective diffusivity D . Motion enters through D , with a separate surface-loss scenario. *Reason:* molecular diffusion alone would not represent circulation; D is anchored only by an order-of-magnitude scaling (Section 2.1) and is tested over a range; vertical mixing is tested separately in Section 9.

2.1 Where the coefficients come from

No temperature record exists to fit, so each coefficient is derived from a textbook relation or tied to a published measurement where one exists, and given a range; where nothing exists (mixing, the exposed fraction, the air and skin temperatures) the value is an assumption and is labelled as one. Surface loss sums natural convection above a hot horizontal surface ($Nu = 0.15 Ra^{1/3}$ for $10^7 < Ra < 10^{11}$, length A/P [5]; here $Ra = 1.9 \times 10^7$), linearized radiation (emissivity 0.96) and evaporation by the Lewis analogy. For open water at 40°C in 22°C air at 50% humidity the parts are 4.7, 6.1 and 24.6 W/(m² K) (0.66 kg/m² h evaporated), 35.4 in total. A bather covers part of the surface; with an assumed exposed fraction of 0.7 the central value is 24.8, and we use 25 within the range 17–37, which spans exposed fractions of 0.5 to 1.0. Evaporation correlations differ by tens of percent, so this is a scenario range, not a measured one.

The shell coefficient is a series resistance: a 5 mm shell (0.19 W/(m K)) with film coefficients of 300 inside and 8 W/(m² K) outside gives 6.5, and 4.7–8.4 for other thicknesses. These are typical engineering values, not measurements of a particular tub. The body coefficient is anchored by a measurement: Menzies et al. report a rectal-temperature rise of $0.9 \pm 0.3^\circ\text{C}$ after 30 minutes in 40°C water to the shoulders [6]. Taking the mean body mass of the study participants, 73 kg, $c = 3470 \text{ J/(kg K)}$ and a mean body rise of 1–2°C, the average heat uptake is 141–281 W, equal to 20–41 W/(m² K) with skin held at 34°C. We use 25, because a fixed skin node overstates the driving difference later in the bath. The mean body rise and the driving difference are assumptions, so this is an order-of-magnitude anchor, not a calibration. Holding the skin at a fixed temperature also ignores the heat capacity of the body, about 37% of the water's here ($73 \text{ kg} \times 3470 \text{ J/(kg K)}$ against 687 kJ/K): uptake is anchored only on its 30-minute average, and its decay with time is not modelled.

Mixing has only a scaling anchor. A mixing-length estimate $D \approx 0.1 u' \ell$, with a velocity scale u' and eddy size ℓ , gives about $1 \times 10^{-4} \text{ m}^2/\text{s}$ for buoyancy-driven flow alone ($u' = 0.02 \text{ m/s}$, $\ell = 0.05 \text{ m}$), 1×10^{-3} for gentle movement of the bather (0.1 m/s, 0.1 m) and 6×10^{-3} for vigorous

movement (0.3 m/s, 0.2 m). We use 0.001 and test 0.0003–0.003. This is a scaling argument, not a measurement, so D stays a scenario.

The 39–41°C window is a preference; immersion studies use 40–42°C water [6], which is not a safety standard. The inlet cell is a point source: the same policy gives a maximum inlet-cell average of 40.55, 40.76 and 41.00°C on the three meshes of Section 11, because a smaller cell mixes less water with the hot inflow. A limit imposed on that cell would test the mesh, not the bath. The ceiling and the spread limit therefore apply outside a jet-mixing zone of 0.15 m radius around the inlet, on the assumption that a bather does not sit in the jet; the floor applies to every cell, which the energy bound of Section 4 needs. With this definition the constrained temperatures agree across meshes (Section 11). An earlier version applied the 41°C ceiling to every cell and tuned it until two meshes agreed; an independent audit showed the limit was not mesh-converged, and that version was replaced.

Table 4. Coefficient anchors and the values used

Coefficient	Relation or anchor	Derived value	Used (range tested)
Surface, W/(m ² K)	Convection + radiation + evaporation [5]	35.4 open; 24.8 at 70%	25 (17–37)
Shell, W/(m ² K)	Series resistance, typical shell values	6.5 (4.7–8.4)	6.5 (4.5–8.5)
Body, W/(m ² K)	Uptake implied by core rise [6]	20–41 at fixed skin	25 (12–40)
Mixing D , m ² /s	Mixing-length scaling	1×10^{-4} to 6×10^{-3}	0.001 (0.0003–0.003)

3 A transparent well-mixed benchmark

Table 5. Symbols

Symbol	Meaning
C	Heat capacity of the water, J/K
H_a, H_b	Air/shell and body conductance, W/K
T_a, T_b, T_{in}	Room, skin and inlet temperature, °C
T_{min}, T_{max}	Lower and upper limit, °C
q, J	Inlet flow, m ³ /s; added water, L
D	Effective mixing diffusivity, m ² /s

Let C be total water heat capacity, H_a the air/shell conductance and H_b the body conductance. A well-mixed overflow has the same temperature as the bath. Integrating the physical energy balance gives

$$C\dot{T} = H_a(T_a - T) + H_b(T_b - T) + \rho c_p q(T_{in} - T) \quad (3)$$

$$C = \rho c_p (LWH - V_b), \quad H_b = h_b A_b \quad (4)$$

$$H_a = h_s f LW + h_w \{LW + 2H(L + W)\} \quad (5)$$

The foam multiplier f equals one without a layer. Setting $q = 0$, define $H = H_a + H_b$ and $T_e = (H_a T_a + H_b T_b)/H$. The cooling solution and first time to reach the lower limit are

$$T(t) = T_e + (T_0 - T_e)e^{-Ht/C} \quad (6)$$

$$t_c = \frac{C}{H} \log \frac{T_0 - T_e}{T_{\min} - T_e} \quad (7)$$

Baseline conductances are $H_a = 37.14$ W/K and $H_b = 28.75$ W/K, and $C = 686,565$ J/K. Thus $T_e = 27.24^\circ\text{C}$ and $t_c = 14.17$ min.

A heat-loss model must approach an environmental equilibrium rather than zero Celsius. This analytic boundary also supplies a direct check on signs and units. The benchmark describes mixing perfectly; the spatial model will test the cost of departing from that assumption.

4 What can be proved about water use?

Let $\ell(T) = H_a(T - T_a) + H_b(T - T_b)$ be the heat loss of the mixed bath, and let the inlet be hotter than every temperature considered. **Proposition 1.** Coasting to the lower limit and then holding it uses the least water, provided the holding rate is within the faucet bound. The proof has four steps.

Step 1, an identity. Dividing the balance by $T_{\text{in}} - T$ and integrating over the horizon gives

$$\int q \, dt = \frac{C}{\rho c_p} \log \frac{T_{\text{in}} - T_0}{T_{\text{in}} - T_f} + \int \frac{\ell(T)}{\rho c_p (T_{\text{in}} - T)} \, dt \quad (8)$$

Step 2, monotonicity. The integrand $\ell(T)/(T_{\text{in}} - T)$ has derivative $[H_a(T_{\text{in}} - T_a) + H_b(T_{\text{in}} - T_b)]/(T_{\text{in}} - T)^2 > 0$, so it increases with T , and the logarithmic term increases with the final temperature. Water use is therefore an increasing functional of the temperature path.

Step 3, the lowest admissible path. Because $q \geq 0$, comparison with the no-flow solution gives $T(t) \geq T_c(t)$, the cooling curve of Section 3, and feasibility requires $T(t) \geq T_{\min}$. The pointwise lowest admissible path is $\max(T_c(t), T_{\min})$: coast until T_c reaches $T_{\min}$, then hold.

Step 4, attainability. Holding $T = T_{\min}$ means $\dot{T} = 0$, which needs the rate

$$q_{\text{hold}} = \frac{\ell(T_{\min})}{\rho c_p (T_{\text{in}} - T_{\min})} \quad (9)$$

When this rate is within the bound, the path is feasible, and by Step 2 it is optimal. ■ For the baseline it adds 16.01 L: wait 14.17 min, then supply 1.012 L/min. This is a proved optimum of the mixed model, not a guarantee for a spatially nonuniform tub.

Proposition 2 (energy lower bound). In the spatial model, suppose every cell satisfies $T_i(t) \geq T_{\min}$. Summing the cell balances gives $\sum_i C_i T_i(t_f) - CT_0 = - \int L \, dt + \rho c_p \int q(T_{\text{in}} - T_{\text{out}}) \, dt$, with L the total loss. Each loss term increases with its cell temperature, and H_a and H_b are the sums of the

cell conductances, so $L(t) \geq \ell(T_{\min})$. The outlet cell satisfies $T_{\text{out}} \geq T_{\min}$, so $T_{\text{in}} - T_{\text{out}} \leq T_{\text{in}} - T_{\min}$, and the left side is at least $C(T_{\min} - T_0)$. Combining these three facts,

$$J \geq \max \left(0, \frac{1000[\ell(T_{\min})t_f - C(T_0 - T_{\min})]}{\rho c_p (T_{\text{in}} - T_{\min})} \right) \quad (10)$$

The resulting 15.41 L is a genuine conditional lower bound, valid for any control, but it is not tight: the initial water is hotter than the floor and loses more heat, and limited mixing adds further cost. Section 7.1 shows how much of the gap an optimized schedule recovers.

5 A conservative model in three dimensions

The rectangular water envelope is divided into $8 \times 4 \times 3$ cells. Each has a capacity C_i and exchanges heat only across common faces. The cells whose centres lie within 0.15 m of the inlet cell form the jet zone (2, 9 and 22 cells on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes); Ω is the rest. Top cells lose heat to the room; bottom and side faces lose heat through the shell. The bather is represented by a smooth, three-dimensional displacement field and a distributed skin contact term.

$$C_i \dot{T}_i = \sum_j g_{ij}(T_j - T_i) + H_{a,i}(T_a - T_i) + H_{b,i}(T_b - T_i) + S_i \quad (11)$$

$$g_{ij} = \rho c_p D \frac{A_{ij}}{d_{ij}} \min(\phi_i, \phi_j), \quad C_i = \rho c_p V_i \quad (12)$$

Here ϕ is fluid fraction, A_{ij} a face area and d_{ij} the center separation. Symmetric g_{ij} guarantees that internal heat exchange cancels when the equations are summed. D is an effective mixing closure; molecular diffusion alone would not represent motion or buoyant circulation.

$$w_i \propto \exp \left[-\frac{1}{2} \sum_{k=1}^3 \left(\frac{x_{ik} - b_k}{s_k} \right)^2 \right], \quad \sum_i w_i = 1 \quad (13)$$

$$V_i = V_{\text{cell}} - V_b w_i, \quad H_{b,i} = h_b A_b w_i \quad (14)$$

Baseline $\mathbf{b} = (0.55L, W/2, 0.45H)$, with widths $\mathbf{s} = (0.40, 0.16, 0.12)$ m. Fluid fraction is $\phi_i = V_i/V_{\text{cell}}$. The center $\mathbf{b}$ and widths $\mathbf{s}$ set the spatial distribution of body effects. Body volume and contact area remain independent inputs: shape is varied through $\mathbf{s}$ at fixed volume and area. This is a homogenized immersed-body representation, not an anatomically resolved obstruction. A cell whose occupied fraction reaches 0.95 is rejected rather than assigned a negative capacity.

A fixed envelope volume is preserved by balancing inlet and overflow. In the baseline the exact displaced-water volume is 164.25 L. Shape scenarios preserve envelope volume when isolating aspect-ratio effects. Volume scenarios deliberately change water depth and hence both storage and side-wall area.

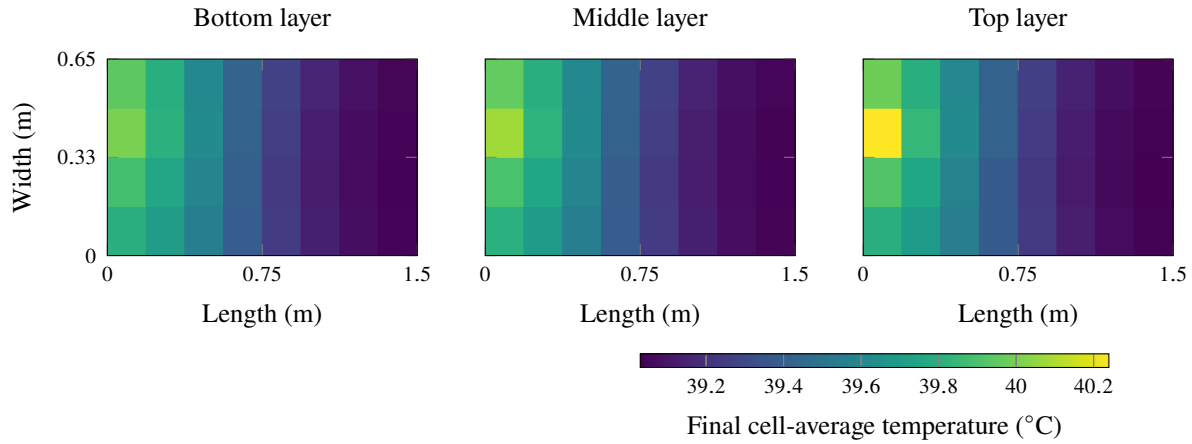


Figure 2. Final temperatures in the three horizontal cell layers. All layers use one color scale; the top layer contains the prescribed inlet-to-overflow stream. Geometry is in metres, and values are cell averages.

5.1 Spatial evidence: the mean is not the whole bath

The heat map is calculated from the same archived trajectory as Figure 3. It shows where the imposed transport path and environmental/body sinks leave temperature differences. The plots are horizontal slices through a three-dimensional network with exchange between layers, not three independent two-dimensional models.

A temperature range summarizes the spread but cannot show its location. The maps make the physical interpretation inspectable: localized replenishment and distributed losses must be balanced through mixing. Every cell contributes to the comfort test; a high mean cannot compensate for a cold region.

The Gaussian body representation changes both storage and exchange distribution. A map does not prove that this homogenized representation captures anatomy, recirculation or buoyancy. Its role is to reveal the actual implications of the declared model, not to create the appearance of a resolved flow simulation. The refinement table separately checks how cell size affects the result.

6 Inlet transport, solver and strategy search

Inlet and overflow are on opposite ends of a surface stream. Every path edge carries the same q : the first cell receives hot water, interior cells receive upstream water and lose the same volume downstream, and the last cell discharges through the overflow.

$$S_1 = \rho c_p q (T_{in} - T_1), \quad S_i = \rho c_p q (T_{i-1} - T_i) \quad (15)$$

$$\sum_i C_i \dot{T}_i = - \sum_i H_{a,i} (T_i - T_a) - \sum_i H_{b,i} (T_i - T_b) + \rho c_p q (T_{in} - T_{out}) \quad (16)$$

This stream deliberately allows hot-water short circuit. Heat can leave at a locally warm outlet before warming a distant cold region. It is an explicit transport assumption, not an inferred flow

field. Alternative inlet placement would require another transport network and calibration.

For a fixed flow, the network is affine linear. Augmenting the state by a constant one permits matrix-exponential propagation, avoiding a large forward-Euler step restriction [4]. A delayed-start policy has two exact constant-control segments.

$$\dot{\mathbf{T}} = \mathbf{A}(q)\mathbf{T} + \mathbf{b}(q), \quad \mathbf{z}(t + \tau) = e^{\tilde{\mathbf{A}}(q)\tau}\mathbf{z}(t) \quad (17)$$

Starts are tested at 0, 120, ..., 1200 seconds. For each, rates from 0 to 3 L/min are bracketed on a 0.2 L/min grid; the first crossing of the lower-temperature requirement is refined by Brent's root method. The search targets $T_{\min} + 0.03^\circ\text{C}$ as numerical reserve, then checks the upper-temperature and spread limits at 5-second output intervals. Starts that already violate the floor are rejected.

This is a reproducible candidate search. Nonmonotone flow responses, untested delays, pulsed inputs, inlet relocation and feedback are not excluded by the calculation. A failed search means no candidate was accepted, not that every possible action is infeasible. The independent checks further examine the chosen trajectory between samples.

7 The spatial result and the mixing penalty

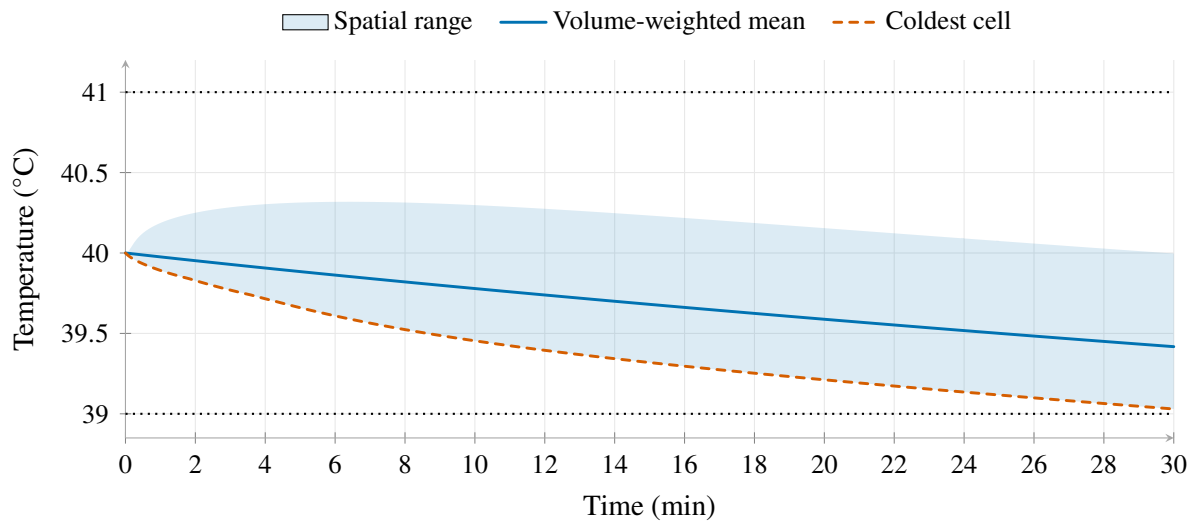


Figure 3. Matrix-exponential cell temperatures under the selected policy. The shaded range covers the coldest cell and the hottest cell outside the inlet jet zone; the mean alone would hide cold and hot locations.

Within the constant-rate family, starting the trickle immediately ranks ahead of delayed starts here, even though waiting is optimal in the ideal mixed model. The distant water needs time to receive heat; larger late rates create a warmer inlet region before they solve the cold-region constraint. This is a concrete consequence of resolving space.

Holding a perfectly mixed bath at 40°C throughout would use 36.22 L. This is a stricter reference service, not a like-for-like optimum. Our accepted 1°C cooling allowance uses less water partly because it provides a different service. The comparable mixed model with the same lower limit

Table 6. Baseline result under the best constant rate

Result	Baseline
Rate / start	0.805 L/min / 0.0 min
Added water	24.14 L
Minimum / maximum	39.030 / 40.318°C
Maximum simultaneous spread	0.967°C
Final volume-weighted mean	39.418°C

uses 16.01 L. The gap also reflects our restricted control family and 0.03°C numerical reserve; it cannot be attributed purely to imperfect mixing.

7.1 A time-varying schedule uses less water

Constant-rate policies are a narrow family. We therefore optimize piecewise-constant inlet flow in K equal segments ($K = 3, 6, 12$) by sequential quadratic programming, minimizing added water under the same all-cell limits (with the 0.03°C reserve), sampled every 5 s, from four starting schedules. The problem is non-convex: each result is the best local optimum found, with no global claim.

Table 7. Constant rate, scheduled flow and the bounds

Policy	Water (L)	Change	Min / max (°C)	Spread (°C)
Best constant rate	24.14	—	39.03 / 40.32	0.97
3 segments	19.75	-18.2%	39.03 / 40.74	1.50
6 segments	19.50	-19.2%	39.03 / 40.73	1.50
12 segments	19.35	-19.9%	39.03 / 40.80	1.50
Perfect-mixing optimum	16.01			
Energy lower bound	15.41			

```

Input: network, limits, K, starting schedules
for each starting schedule x0:
    minimize sum(x)*segment_time over 0 <= x <= 3 L/min
    subject to min_i T_i(t) >= Tmin+reserve, max_{i in Omega} T_i(t) <= Tmax,
               spread over Omega <= span (every 5 s)
    replay the best x at 5 s; keep it only if all margins >= 0

```

The 12-segment schedule is off for the first 7.5 minutes and off for the last 10.0 minutes, with a peak of 2.81 L/min between. It adds 19.35 L, 19.9% below the constant rate, and narrows the gap to the energy lower bound from 8.7 L to 3.9 L. Refining from 3 to 12 segments gains only 2.0%: the saving comes from the shape. The optimum touches the lower limit and the spread limit (1.50°C) and stays 0.20°C below the upper limit (40.80°C).

We have not isolated why the best schedule stops at the end. One candidate explanation, an inference rather than a model result, is that hot water added late leaves through the overflow before

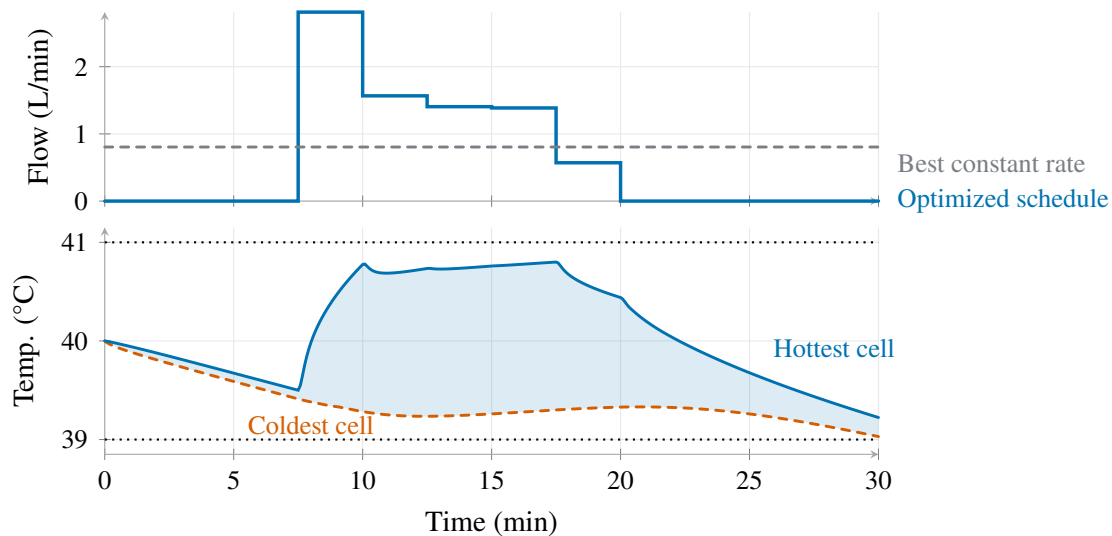


Figure 4. Best 12-segment schedule against the best constant rate (top) and the range of cell temperatures outside the inlet jet zone (bottom); dotted lines mark the limits.

it warms the remote cells, while stored heat carries the final minutes.

The schedule is not an artefact of the mesh. Replayed on the $8 \times 4 \times 3$, $12 \times 6 \times 4$ and $16 \times 8 \times 6$ meshes it gives spreads of 1.500, 1.498 and 1.524°C and maxima of 40.80, 40.79 and 40.82°C outside the jet zone. Re-optimizing the 6-segment schedule on the $12 \times 6 \times 4$ mesh gives 19.44 L against 19.50 L, with flows (0.00, 0.78, 1.72, 1.40, 0.00, 0.00) against (0.00, 0.81, 1.70, 1.39, 0.00, 0.00) L/min: the same shape, off at both ends. The result comes from two starts of the same search, so it supports the shape, not a global optimum.

7.2 What does staying close to the start temperature cost?

The task asks for a bath close to its initial temperature without wasting much water, which is a trade-off rather than a single optimum. Figure 5 repeats the search while the allowed fall below 40°C changes from 0.25 to 3°C; the 1°C case is the baseline. With a fall of 2.5°C or more the stored heat suffices and no water is added. At 2°C the best constant rate adds 1.9 L, at 1.5°C 12.9 L, at 1°C 24.1 L and at 0.75°C 29.8 L. Between 2°C and 0.75°C each further 0.5°C of tolerance is therefore worth roughly 11 L, almost linearly. At 0.5°C the best constant rate adds 36.3 L; for a fall of 0.25°C no constant-rate policy was accepted, while the perfectly mixed bath would still need 30.7 L.

The curve is the practical answer to “how close is close enough”: below a 2.5°C fall every degree of tolerance bought back costs about the same, so the tolerance is worth choosing deliberately. The perfectly mixed curve lies below the spatial one at every tolerance, and the gap widens as the tolerance tightens, from 1.5 L at 2°C to 9.2 L at 0.75°C: the stricter the comfort requirement, the more uneven mixing costs. Only constant-rate policies are searched here; scheduled flow (Section 7.1) lowers the baseline point by about 20% and would shift the brown curve down by a similar fraction, an expectation that was not recomputed at the other tolerances.

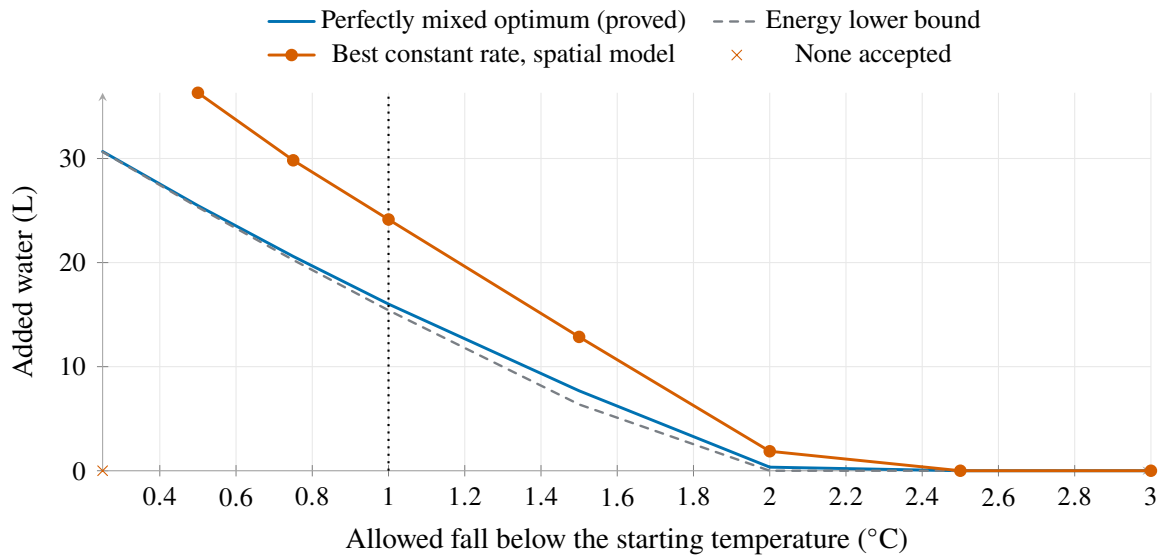


Figure 5. Added water against the allowed temperature fall. The mixed optimum and energy bound are closed forms; markers on the brown curve are searched constant-rate policies, and the dotted line is the baseline.

7.3 Can a user follow the schedule?

A schedule that a person cannot follow is of little use, so we perturb the tap instead of the physics. The optimized schedule touches the spread limit, so its slack to the stated limits is zero and any error that raises the spread crosses it; what matters is by how much. Scaling every segment by 0.9 or 1.1 crosses a limit by 0.076 and 0.126°C, and by 0.8 or 1.2 by 0.183 and 0.249°C. With independent errors of 10% in each segment the median draw misses a limit by 0.070°C and the 5th-percentile draw by 0.217°C (200 draws, fixed seed); at 20% they miss by 0.147 and 0.373°C. The distances are small next to the 1.5°C limit, but they lie on the wrong side.

The error is not symmetric. The most sensitive segment is segment 4 (2.81 L/min), where raising the flow by 20% breaks the limits by 0.245°C and lowering it by 20% by 0.047°C: extra hot water overheats the inlet region and widens the spread. Over-delivery in a high-flow segment is the error to guard against. 1 of the 5 active segments tolerates a 20% error in either direction.

The price of a margin is measured on a 6-segment version. Requiring all three limits to hold with a buffer b gives 19.50 L at $b = 0$, 21.48 L (+10%) at 0.1°C, 23.81 L (+22%) at 0.2°C and 26.83 L (+38%) at 0.3°C. With the 0.1°C buffer, 74% of 200 draws with 10% segment errors stay within the limits. A user who cannot hold the tap to within a few percent should buy the 0.1°C buffer or accept the constant trickle, which does not sit on the spread limit.

8 Separate geometry, size and body effects

At equal volume, the wide, shallow tub has a larger exposed top area and a changed diffusion length; it has no accepted candidate. The deep, narrow tub needs 17.48 L against 24.14 L, because a deep,

Table 8. Geometry, size and body scenarios

Change from baseline	Water (L)	Interpretation
Shallow Wide	No candidate	Same envelope volume
Deep Narrow	17.48	Same envelope volume
Small Bath	27.05	Lower water depth
Large Bath	20.39	Higher water depth
Deep Bath	13.05	Depth 0.35 m, like a bath filled to the shoulders
Larger Body	28.11	More displacement/contact
Long Body	23.85	Shape only; fixed volume/area
Warmer Skin	21.08	Skin at 36°C
Cooler Skin	27.18	Skin at 32°C

narrow envelope reduces top-area heat loss, although a real deep bath may stratify, which Section 9 tests with a separate vertical-mixing scenario. Geometry is therefore more than an interchangeable cooling coefficient.

Changing depth at fixed length and width changes thermal storage and shell area. Greater water volume slows cooling, which can lower added water over a short fixed horizon: the large bath needs 20.39 L, while the small bath needs 27.05 L. The 0.23 m baseline depth is shallow, and its surface-to-volume ratio is about 1.7 times that of a bath filled to the shoulders; a 0.35 m depth needs 13.05 L, so the baseline overstates the surface-loss share of a deeper bath. That does not mean a larger bath conserves total water: filling it initially uses more. J counts replenishment only, and this distinction prevents a misleading conservation claim.

The larger-body scenario simultaneously displaces more water and increases assumed contact area; it is a combined size scenario and needs 28.11 L. The long-body scenario isolates the spatial distribution by keeping total body volume and contact area fixed. Warmer skin reduces the temperature difference driving body heat loss. These changes quantify conditional dependence; they do not establish measured human heat transfer.

To transfer the model to a real tub, measure water volume after entry, submerged contact geometry, temperatures at several depths and distances, and no-inlet cooling. Shape and temperature should not be inferred from a single mean cooling curve.

9 Motion and a bubble layer can change the policy

Increasing D reduces the gradient created by the localized inlet. The strong-mixing scenario uses 19.40 L against 24.14 L, but motion may also increase heat loss. When D is increased together with a 20% increase in the surface coefficient (30 instead of 25 W/(m² K)), the need rises to 23.04 L and most of the benefit disappears. The comparison deliberately separates transport improvement from its possible boundary cost.

Stratification is tested separately. A hot inlet layer is lighter than the water below, and the stable

Table 9. Mixing, surface, comfort and supply scenarios

Scenario	Added water (L)	Search outcome
Weak Mixing	—	None accepted
Strong Mixing	19.40	Accepted
Moving With Added Surface Loss	23.04	Accepted
Stratified	24.42	Accepted
Convective Surface Layer	24.10	Accepted
Foam	10.37	Accepted
Tight Comfort	36.32	Accepted
Loose Comfort	1.87	Accepted
Low Loss	8.60	Accepted
High Loss	—	None accepted
Cool Supply	44.80	Accepted

density gradient suppresses vertical mixing. With a cell height of 0.077 m, a vertical temperature difference of 1 K, a velocity scale of 0.02 m/s and an expansion coefficient of about 3.8×10^{-4} per K, the gradient Richardson number is $Ri = g\beta \Delta T \ell / u^2 \approx 0.7$. The Munk–Anderson stability function for scalars, $(1 + 3.33 Ri)^{-3/2}$ [8], then reduces vertical diffusivity to about 0.15 of its neutral value. We test a vertical-to-horizontal ratio of 0.2. The best constant rate becomes 24.42 L against 24.14 L, but the maximum spread rises from 0.97 to 1.07°C against the 1.50°C limit. Stratification therefore costs little water here and uses 19% of the remaining uniformity margin; the ratio is an order-of-magnitude scenario and the Richardson estimate uses assumed velocity and temperature scales. The stable layer is an upper-bound picture: evaporative cooling at the surface can instead drive convection and raise vertical mixing, so a vertical-to-horizontal ratio of 3 is also tested, and it needs 24.10 L.

The foam scenario multiplies only the effective surface coefficient by 0.4. With all other inputs fixed, the best constant rate falls to 10.37 L. The assumed 60% reduction is a scenario, not an experimentally established property of bubble-bath additive. If the layer breaks up or motion raises evaporation, this result must be recomputed.

No accepted candidate exists for: Weak Mixing, Shallow Wide, High Loss. They are reported as such, not hidden in a favorable average. In the weak-mixing, high-loss and wide-shallow cases, flows large enough to keep every cell above the floor already push the spread above its 1.5°C limit (checked at constant flows from 0.8 to 2.5 L/min), so the stated limits cannot be met together by a constant flow there; this concerns the stated limits, not what a bath can physically do. Their outcomes identify where the restricted strategy must change. Potential responses include improved circulation, a different inlet path, a shorter bath, or a different comfort tolerance. The paper does not certify which response is safe or optimal without corresponding physical evidence.

10 Validation: independent evidence and remaining error

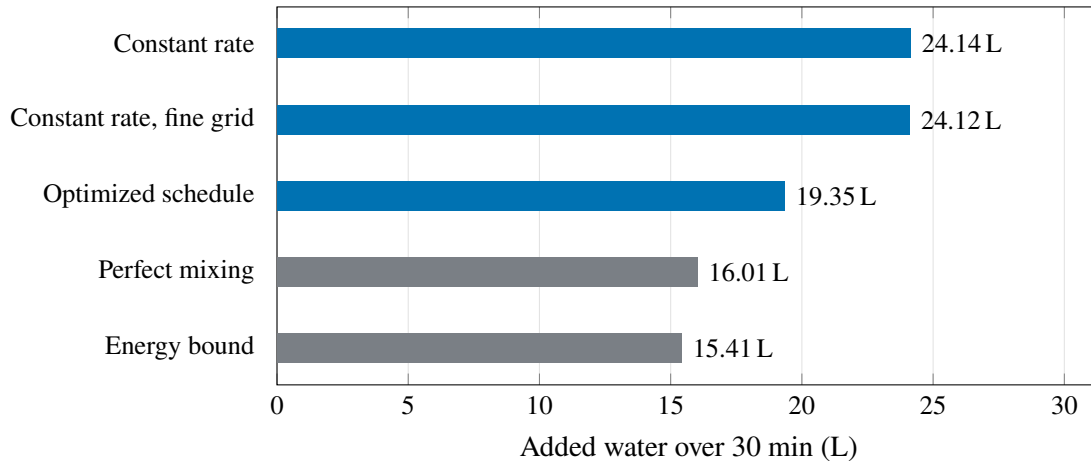


Figure 6. Water-use comparison under the same lower limit. Blue bars are simulated policies; grey bars are analytic bounds, and the energy bound is conditional on the network assumptions.

All 17 recorded checks pass. Independent heat-flow arithmetic sums environmental losses, body exchange and local-temperature overflow; the largest instantaneous residual is 2.46×10^{-10} W. An adaptive RK45 integration with independently assembled right-hand side agrees with the archived matrix-exponential trajectory within $3.12 \times 10^{-07}^\circ\text{C}$. This tests numerical implementation, not a second physical model.

A constructed uniform-loss problem reproduces its analytic exponential cooling solution. Internal exchange cancels globally, and the stream has exactly one overflow sink. Positive off-diagonal dynamics and nonpositive row sums support physical bounds and a contractive derivative estimate for each constant-flow segment.

One-second sampling is supplemented by a derivative bound: if $|dT_i/dt| \leq L$, a point lies at most 0.5 seconds from a sample, so temperatures differ by at most $L/2$ and spread by at most L . With a numerical allowance, the resulting lower envelope is 38.9989°C , upper 40.3496°C , and spread upper bound 1.0291°C . This is a conditional floating-point envelope, not interval-arithmetic certification.

The optimized schedule is replayed with the independently assembled right-hand side and RK45 at one-second samples. It agrees with the matrix-exponential trajectory within $3.2 \times 10^{-07}^\circ\text{C}$ and satisfies the limits (minimum 39.030°C , maximum 40.800°C , spread 1.501°C , each within the 0.002°C sampling tolerance); its 19.35 L exceeds the 15.41 L energy bound, as it must. These two checks cover this one schedule and are separate from the 17 model checks.

11 Resolution, uncertainty and transfer to a real bath

The same constant-rate policy is replayed on three meshes. The ceiling and spread quantities outside the jet zone agree within 0.015°C and 0.013°C , so the comfort limits do not depend on the mesh.

Table 10. Mesh replay: constrained temperatures

Mesh (cells)	Max outside jet zone	Spread outside zone	Inlet-cell max
$8 \times 4 \times 3$ (96)	40.318°C	0.967°C	40.553°C
$12 \times 6 \times 4$ (288)	40.315°C	0.960°C	40.757°C
$16 \times 8 \times 6$ (768)	40.330°C	0.973°C	40.995°C

The inlet-cell maximum rises from 40.55 to 41.00°C with refinement; that is why the cell is excluded rather than constrained. The selected water amounts on the $8 \times 4 \times 3$ and $12 \times 6 \times 4$ meshes are 24.142 and 24.117 L (0.11% apart). Three meshes are a diagnostic, not an asymptotic convergence-order study.

Parameter uncertainty concerns heat loss, body exchange and mixing. Structural uncertainty concerns the prescribed surface stream, fixed skin reservoir, homogenized body and constant effective D . These are different errors. The scenario table explores parameter dependence and one coupled motion effect; it is not a probability distribution or confidence interval.

Range analysis. Section 2 gives ranges for six inputs. 64 scrambled Sobol points cover them (D log-uniform), and the constant-rate search runs at each. 41 points have an accepted candidate. Among those, the required water is 12.5, 17.0, 22.0, 26.9 and 30.9 L at the 5th, 25th, 50th, 75th and 95th percentiles. The remaining 23 points have none, mostly because of weak mixing (below). The baseline of 24.1 L sits in the upper half of this distribution.

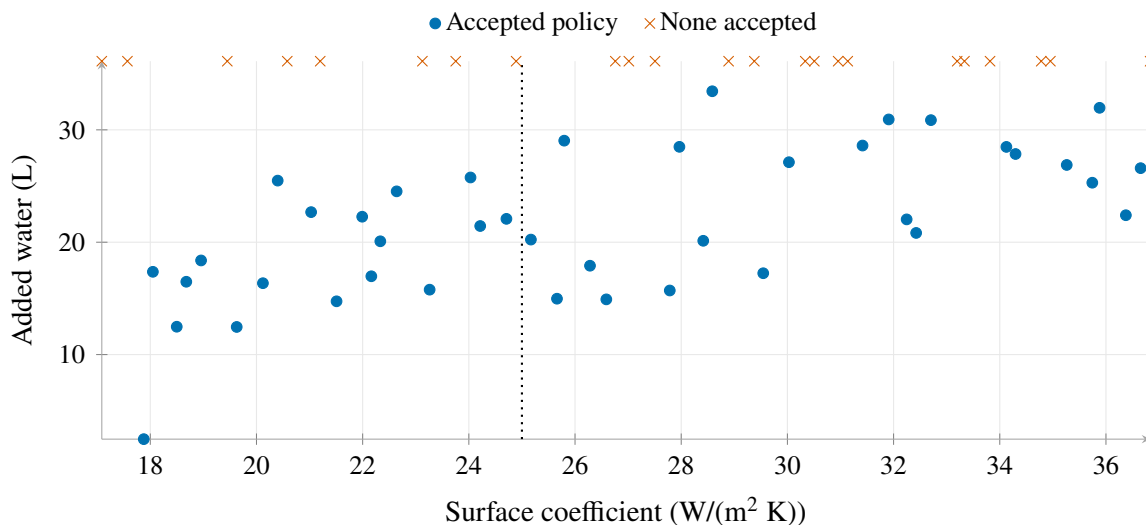


Figure 7. Required water against the surface coefficient for each Sobol draw. Crosses at the top mark draws with no accepted policy; the vertical line is the baseline.

Spearman rank correlations with water, over accepted draws: surface coefficient +0.63, room temperature -0.47, body coefficient +0.47, skin temperature -0.27, shell coefficient +0.15, mixing diffusivity -0.15.

The correlations use accepted draws only, so they say nothing about which draws have an ac-

cepted policy at all. That is decided mainly by the mixing diffusivity: its median is 1.4×10^{-03} m²/s for draws with an accepted policy and 4.5×10^{-04} for draws without (Mann–Whitney $p < 1 \times 10^{-4}$), while the other five inputs show no association (smallest $p = 0.47$). Weak mixing leaves the far water too cold unless the flow is raised so far that the inlet region breaks the spread limit. The draws are uniform over ranges that we chose; they are not a probability distribution for any real tub. The main message: once a policy exists, the loss coefficients and the room temperature decide how much water it needs, and the mixing closure decides whether one exists, which is why Section 2 ties the coefficients to outside evidence and Section 9 reports weak mixing separately.

A calibration plan should begin with an inlet-free cooling experiment, recording water volume and room conditions. Multiple temperature probes distinguish aggregate heat loss from mixing. A separate inlet experiment records flow and inlet/outlet temperatures, allowing overflow energy to be checked. Motion and foam need matched trials because both can alter transfer coefficients. Reserve an entire experiment for prediction checks after fitting. A single mean-temperature series cannot separate the surface, shell and body coefficients, so only calibrated and independently tested parameters would support advice for a particular bath.

12 Conclusions and a policy with clear scope

The model supports four linked findings. First, tolerating a modest temperature decline lets stored heat replace some inlet water. Second, spatial transport changes the ideal mixed recommendation: the best constant rate starts at once, and a schedule that withholds water at both ends needs 20% less. Third, the water required is dominated by the surface and body loss coefficients: from 12 to 31 L across the scenario ranges. Fourth, improving mixing helps only to the extent that it does not create offsetting boundary loss or hot-water short circuit.

For the stated baseline, the best constant trickle is 0.805 L/min for 24.14 L over 30 minutes; the 12-segment schedule uses 19.35 L. These are the best accepted results of the documented searches, with all-cell constraints and independent numerical checks. Neither is a universal faucet prescription or a proved global optimum over all time-varying controls.

The spatial model earns its complexity by explaining a disagreement with a proved simple benchmark and by rejecting strategies that a mean-only calculation would accept. Conservative accounting, coefficients with stated ranges and labelled assumptions, a feasible-policy versus lower-bound gap, and an explicit refinement check make the recommendation inspectable.

Strengths. The conservation laws hold exactly and are checked by an independently assembled integrator; the benchmark and the lower bound are proved, so the spatial result is bracketed; the coefficients carry sources, ranges and a sensitivity analysis; the optimized schedule shrinks the gap to the bound from 8.7 to 3.9 L.

Weaknesses. The surface stream and the effective D stand in for flow physics; the body coefficient rests on an assumed mean body rise and skin gradient; the schedule is a local optimum; the 0.15 m jet zone outside which the ceiling and spread apply is an assumption; and no bath experiment was run.

The limitations are consequential. Three-dimensional geometry is represented, but fluid momentum, buoyancy, jet entrainment, free-surface motion and detailed body anatomy are not solved. No bath experiment was conducted. Temperature limits express a preference assumption. Coarse cell averages cannot establish burn safety near the inlet, and the shape scenarios do not resolve stable vertical layers. The shell's own heat capacity (a few percent of the water's for a thin plastic or enamelled steel tub, more for cast iron or ceramic) is ignored.

Before using a numerical rate in practice, identify the real tub's cooling and mixing behavior. Until then, the defensible transferable advice is to avoid unnecessary replenishment, distinguish cold-region temperature from the mean, improve distribution before increasing flow, and reconsider a strategy when geometry or motion changes. The following page translates these principles without requiring the user to interpret the equations.

13 A warmer bath, with less replacement water

A guide for the person in the bathtub

Decide what “warm enough” means. A bath need not stay at exactly its starting temperature to remain acceptable. In our example, allowing a 1°C fall still needs about 24 L of replacement water, while allowing a 2°C fall needs only 1.9 L. Your actual comfort and health needs must determine the acceptable range.

Distribute the warmth before turning up the tap. The water near the faucet can warm while the far end remains cool. Gentle movement can help redistribute heat. More vigorous motion is not automatically better: it may also increase heat loss from the surface.

Run the tap in the middle, not at the ends. For the tub and mixing conditions analyzed here, the least-water schedule adds no hot water for the first 7.5 minutes or the last 10 minutes, with most of the water in between; a constant trickle from the start used 25% more. The exact minutes change with the tub, but stopping before the end held on every mesh we tried. In a bath that mixes very quickly, waiting until it has cooled slightly can save more. There is no single schedule that works for every tub.

Do not follow a clever schedule too exactly, or too generously. The least-water schedule sits on the edge of what is comfortable: a tap that runs 10% hot or cold crosses the edge by about a tenth of a degree, and extra hot water does more harm than too little. Keep a small margin; in our example it costs about 10% more water.

Check more than one location. A comfortable mean temperature can hide a cold far corner or a hot inlet region. Check the bath away from the faucet and at different depths. Do not use a model’s cell-average temperature as a reason to contact a hot jet.

Remember where the added water goes. Once the tub is full, extra water leaves through the overflow. Hot water that runs along the surface to the overflow may escape before its warmth reaches you. Increasing the stream can waste heat without fixing unevenness.

A bubble layer may help, but the amount is uncertain. An intact insulating surface layer can reduce heat loss. Our calculation explores such a layer; it does not measure the effect of a particular additive. If the layer disappears or the bath behaves differently, reassess rather than relying on the example’s saving.

The practical rule: use the bath’s stored warmth, keep temperature distributed, and add only as much water as your actual conditions require. The numerical example explains these tradeoffs; it is not a tested bathing or safety standard.

14 References and reproducible algorithm

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Algorithm and computational record

1. Form cell volumes and capacities; reject excessive body occupancy. Assemble symmetric face exchange, environmental/body conductances and the conservative stream. 2. Compute the mixed closed forms and energy lower bound. 3. Evaluate delayed constant-flow candidates with matrix exponentials and refine the first bracketed lower-temperature crossing. Reject ceiling or spread violations outside the jet zone. 4. Rank accepted candidates by added litres. 5. Independently integrate and check balances, analytic limits, parameter arithmetic and constraints; replay a finer mesh and perturb physical scenarios. 6. Optimize piecewise-constant schedules (multi-start SQP) and replay the best independently. 7. Rerun the constant-rate search at Sobol draws over the literature ranges. Independent RK45 checks use relative tolerance 2×10^{-9} , absolute $2 \times 10^{-10}^\circ\text{C}$ and steps of at most 5 s. AI assistance is acknowledged in [7].

Appendix A. Scenario definitions

Each row changes only the listed baseline inputs. All unspecified inputs remain as stated in Section 2; geometry and conductance are then rebuilt. The equal-volume aspect-ratio cases retain the envelope volume of 0.22425 m^3 . Widths s describe a smooth displacement distribution, not measured anatomical semiaxes.

Table 11. Scenario definitions

Scenario	Changed input(s)
Weak Mixing	$D \text{ (m}^2/\text{s)} = 0.0003$
Strong Mixing	$D \text{ (m}^2/\text{s)} = 0.003$
Moving With Added Surface Loss	$D \text{ (m}^2/\text{s)} = 0.003$; surface coefficient ($\text{W}/\text{m}^2\text{K}$) = 30
Loose Comfort	lower limit ($^{\circ}\text{C}$) = 38
Tight Comfort	lower limit ($^{\circ}\text{C}$) = 39.5
Foam	surface multiplier = 0.4
Shallow Wide	length (m) = 1.7; width (m) = 0.75; depth (m) = 0.175882
Deep Narrow	length (m) = 1.3; width (m) = 0.6; depth (m) = 0.2875
Small Bath	depth (m) = 0.2
Large Bath	depth (m) = 0.27

Accepted candidates are independently replayed on their scenario network at intervals of no more than five seconds. These scenario checks are sampled checks; the stronger one-second continuous-time envelope is reported for the selected baseline policy only. “None accepted” refers to the documented finite search and is not an infeasibility theorem.

Appendix A. Scenario definitions (continued)

Each row changes only the listed baseline inputs. All unspecified inputs remain as stated in Section 2; geometry and conductance are then rebuilt. The equal-volume aspect-ratio cases retain the envelope volume of 0.22425 m^3 . Widths s describe a smooth displacement distribution, not measured anatomical semi-axes.

Table 12. Scenario definitions (continued)

Scenario	Changed input(s)
Deep Bath	depth (m) = 0.35
Larger Body	body displacement (m^3) = 0.075; body contact area (m^2) = 1.35
Long Body	body widths s (m) = [0.48, 0.12, 0.1]
Warmer Skin	skin temperature ($^{\circ}\text{C}$) = 36
Cooler Skin	skin temperature ($^{\circ}\text{C}$) = 32
High Loss	surface coefficient ($\text{W}/\text{m}^2\text{K}$) = 36; wall coefficient ($\text{W}/\text{m}^2\text{K}$) = 8.5; body coefficient ($\text{W}/\text{m}^2\text{K}$) = 40
Low Loss	surface coefficient ($\text{W}/\text{m}^2\text{K}$) = 17; wall coefficient ($\text{W}/\text{m}^2\text{K}$) = 4.5; body coefficient ($\text{W}/\text{m}^2\text{K}$) = 12
Stratified	vertical/horizontal D ratio = 0.2
Convective Surface Layer	vertical/horizontal D ratio = 3
Cool Supply	inlet temperature ($^{\circ}\text{C}$) = 45

Accepted candidates are independently replayed on their scenario network at intervals of no more than five seconds. These scenario checks are sampled checks; the stronger one-second continuous-time envelope is reported for the selected baseline policy only. “None accepted” refers to the documented finite search and is not an infeasibility theorem.

Report on Use of AI

Tool and scope

OpenAI Codex, a GPT-6-based assistant, was used on 7 October 2026. The later revision (textbook-correlation coefficients, schedule optimization, range analysis, a mesh-convergence audit and re-building of this report) used Anthropic's Claude, Sonnet 5.5, through Claude Code on the same date. Structure and presentation were informed by five Outstanding papers on other problems; none on this problem was read. The assistant once read COMAP's published judges' commentary on this problem during development and then removed every change it had prompted. The assistant selected the historical problem, located and read sources, formulated the mixed and spatial models, wrote and revised Python code, designed checks, interpreted numerical runs, generated figures, and drafted and typeset the English report. An assistant sub-agent wrote the independent-RHS validation routine; this is AI-assisted code review, not review by another human.

Task

The task was to select a classic historical MCM problem and build a rigorous, professional case from it. The conversation wording is not reproduced in this report or in the development record.

Outputs, corrections and verification

AI-produced outputs are the model, validation and report-building sources; the mathematical derivations, scenario tables, figures and text in this document. Numerical values came from actual Python runs. Citations [1]–[4] were checked against the linked sources; [5] and [8] were not opened (their relations were checked against web summaries and a turbulence-model source) and [6] was read in full from the open-access author manuscript. The first baseline coefficients were replaced after anchoring. An independent audit found that the inlet-cell ceiling and the spread did not converge with the mesh; the limits were redefined outside a jet zone and every result was recomputed.

Record limitations

A full exported interaction transcript and every intermediate AI output were not available in this artifact. The task description, tool identification, scope, code artifacts and numerical receipts are retained in the accompanying development record; this report does not invent a transcript. No independent human review or physical bath experiment is claimed.

Python, NumPy and SciPy performed the mathematics; the report was typeset in XeLaTeX, with figures drawn by pgfplots and TikZ from the archived numerical record. These deterministic tools do not establish empirical validity. The service model/build beyond the available GPT-6-based identifier was not independently verified.